



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -1 (Paper-1)

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Computer Fundamental

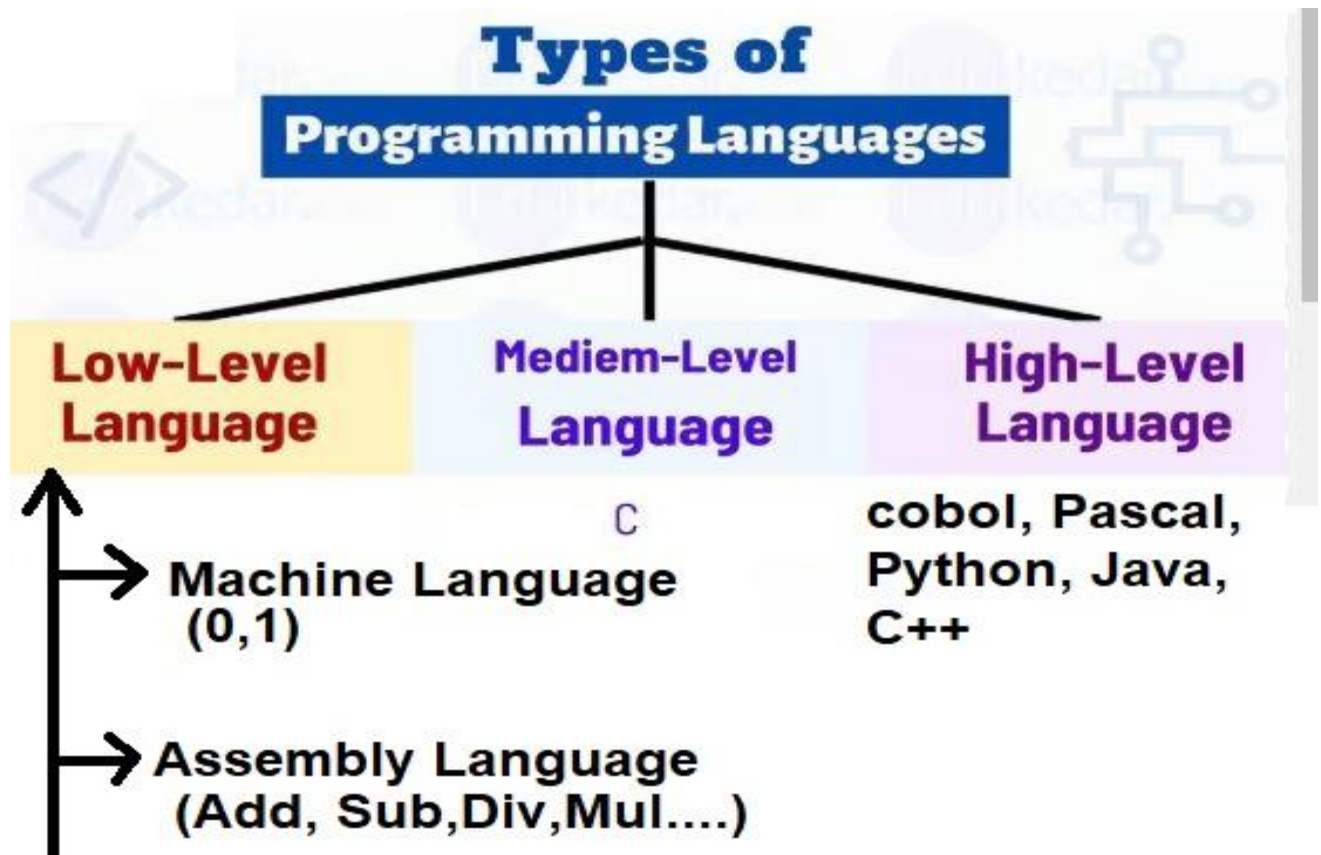
COMPUTER SOFTWARE

❖ **Programming Language**

- A computer can only do, what a programmer writes.
- Now as we know only human languages and computer knows only machine language, we need some media through which we can communicate with the computer.
- Languages are tools/media human can use to communicate with the hardware of a computer system.
- Each language has a systematic method (व्यवस्थित विधि) of using symbols of that language.e.g.- In English, this method is given by the rules of grammar.
- Similarly, the symbols of particular one computer language must also be used as per set of rules which are known as the “Syntax” of that language.

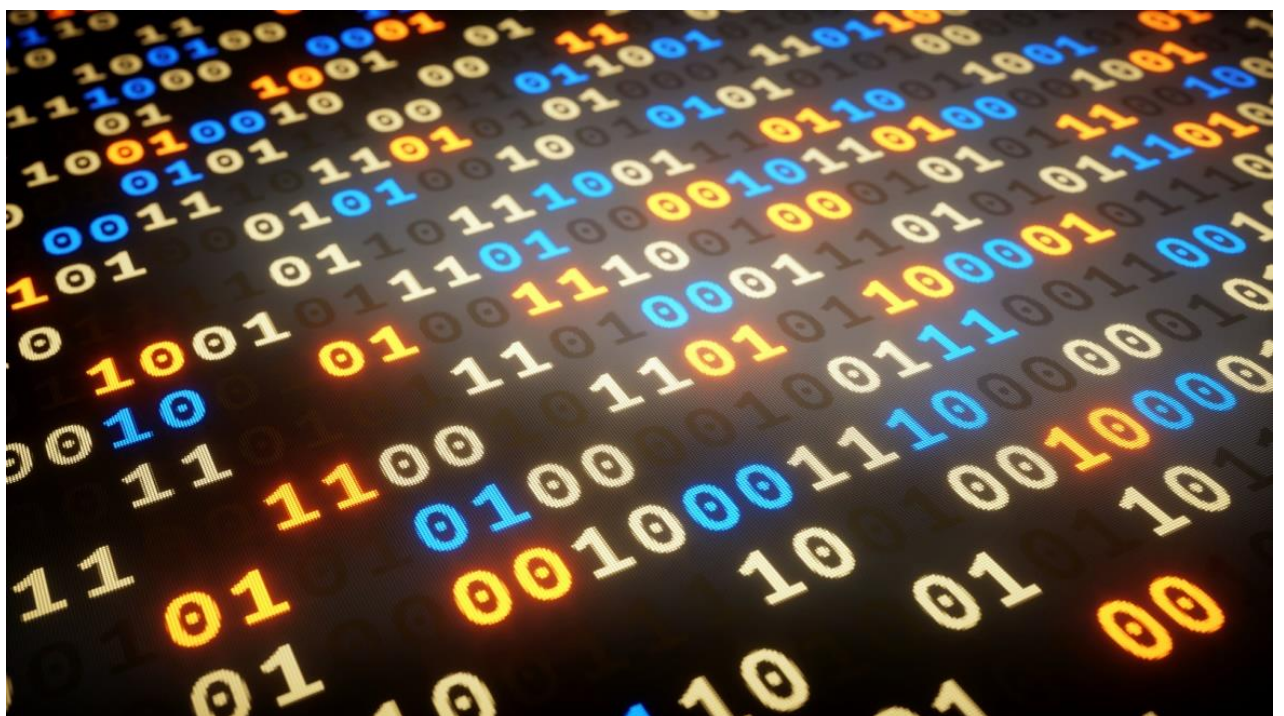
➤ **Computer Languages can be classified into three categories.**

- i) **Low Level Languages**
- ii) **Medium Level Languages**
- iii) **High Level Language**



1. Low Level Language

- Low level languages are made according to the internal working of the computer and the speed of execution of programs written in such languages is high, because the computer can follow its instructions directly.
- Its two main examples are- machine language & Assembly language.



i) Machine language

- Computer programs are written using many different computer Languages(c,c++,java,HTML) but the language which is understood by the computer without translating program is called machine language.
- Machine language is normally written as string of binary 1s and 0s.

Advantages

- ✓ Programs can be executed immediately, because it doesn't require any translation.
- ✓ Now extra storage space is needed.
- ✓ Programmer has complete control over the performance of the hardware.

Disadvantage

- ✓ Difficult to program
- ✓ Tedious (थकाउ)/ Boring to program
- ✓ Difficult to modify
- ✓ Time consuming to code
- ✓ Many Error detect
- ✓ Operation codes have to be memorized
- ✓ Assignment of memory is done by programmer
- ✓ Time consuming for development
- ✓ Programs development are machine dependent
- ✓ Preparation of programs was slow and costly.

ii) Assembly Language

- Assembly language is a language which allows instruction & storage location to be represented by letters & symbols, instead of number.
- A program written in an assembly language is called assembly language program or symbolic program.
- Assembly language was introduced in 1952.
- Machine language was tedious to code and errors were expected to arise in bulk. To solve these problems mnemonic codes (HQ for headquarters, GDC for govt. degree college, Govt. for Government) and symbolic addresses (Names for memory locations) were developed.

- This language translation is actually done by a special translating program, this is called Assembler.

Advantage

- ✓ Easier to use, code and understand.
- ✓ Easier to correct error.
- ✓ Easier to modify.
- ✓ Efficiency of machine language.
- ✓ Can use Macros (Macro is a bunch of instruction referred as a single name-e.g.- ADD)

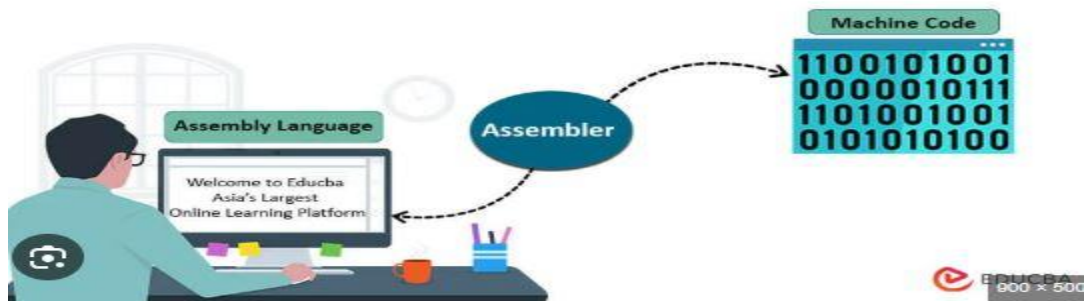
Disadvantage

- ✓ Machine depended.
- ✓ Programs have to be translated before execution.
- ✓ Translation of programs takes up time.
- ✓ Knowledge of hardware is required.
- ✓ Additional storage area needed for the source programs and object code.
- Examples of Assembly Language - Microsoft Assembly Language (MASM), Turbo Assembler

```
begin
PAUSE
MOV Rsw Rsrc1
MOV Rsw R7seg
PAUSE
MOV Rsw Rsrc2
MOV Rsw R7seg
ADD
PAUSE
MOV Rdest R7seg
end
```

```
0000
F000
0079
0078
F000
007A
0078
1100
F000
00B8
```

Assembly Language



2. Midium Level Language

- languages act as a bridge between low level and high level languages.
- C language is called a middle level language because it has the qualities of both high level and low level languages.

The screenshot shows a text editor window with a menu bar (File, Edit, Search, Run, Compile, Debug, Project, Options, Window, Help) and a title bar (MONTECAR.C). The code is as follows:

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

void main()
{
    int j,count=0,limit,i=0,head,tail;
    float y,answer,h=2.5;
    clrscr();
    printf("enter the limit amount");
    scanf("%d",&limit);

    while(i<=limit)
    {
        i++;
        head=0;
        tail=0;
        for(j=0;j<10;j++)
        {
            y=(float)((rand()% 65535)/65535.0f)*10;
            printf("%f\n",y);
        }
    }
}
```

3. High Level Language

- The machine language & assembly language requires a good knowledge of internal structure of computer.
- The high level languages machine independent so it can be easily ported & executed on any computer.
- The high level language programs do not require any knowledge of internal structure of computer so the programmer concentrate on the logic of problem rather(अपेक्षा) than internal structure of computer.
- It enables the programmer to write instructions using English words & familiar mathematical symbols & expression so the program makes easier to code & understand.

- It requires a translator program to convert high level program into machine language, this device is called Compiler.

Advantages:

- ✓ Machine independent.
- ✓ Easier to learn, use and understand.
- ✓ Easier to correct error.
- ✓ Easier to maintain.
- ✓ Less time & efforts(प्रयास).
- ✓ .Program preparation cost is low.
- ✓ Few errors.

Disadvantages:

- ✓ Slower execution
- ✓ Lower efficiency(क्षमता).
- ✓ Require more storage space

e.g – C++, Java, python, Javascript,....

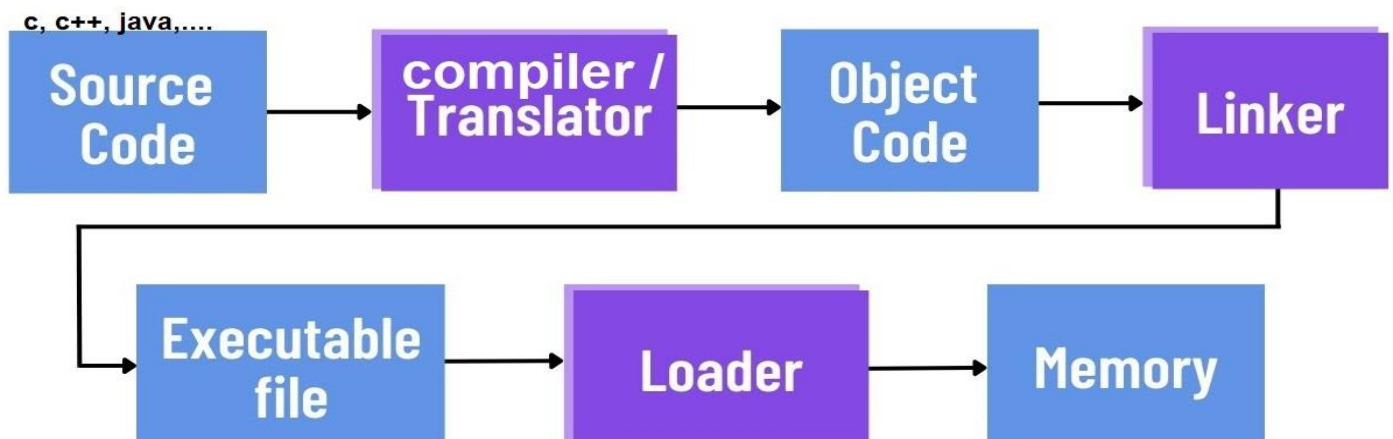
```
new.py > ...
1  number_1 = input('Enter your first number: ')
2  operator = input('operator: ')
3  number_2 = input('Enter your second number: ')
4
5  if (operator == '+'):
6      print("Rusult:", number_1+number_2)
7  elif(operator == '-'):
8      print("Rusult:", number_1-number_2)
9  elif(operator == '*'):
10     print("Rusult:", number_1*number_2)
11 elif(operator == '/'):
12     print("Rusult:", number_1/number_2)
13 else:
14     print('Try again')
```

❖ Difference B/w High Level & Low Level Language

S.NO	HIGH LEVEL LANGUAGE	LOW LEVEL LANGUAGE
1.	It is programmer friendly language.	It is a machine friendly language.
2.	High level language is high memory efficient.	Low level language is less memory efficient.
3.	It is easy to understand.	It is tough to understand.
4.	It is simple to debug.	It is complex to debug comparatively.
5.	It is simple to maintain.	It is complex to maintain comparatively.
6.	It is portable.	It is non-portable.
7.	It can run on any platform.	It is machine-dependent.
8.	It needs compiler or interpreter for translation.	It needs assembler for translation.
9.	It is used widely for programming.	It is not commonly used now-a-days in programming.
10.	e.g. - C,C++,JAVA, BASIC	e.g - Machine language, Assembly language

❖ Linker

- When a program written in a High level language(HLL) and compiler/Translate into machine language, the output thus obtained is called An Object file / Object program / Object Code(.obj / .O).
- After which a program linker link combines all object files and make an actual executable file(.exe).



❖ Loder

- A loder is a type of System software that lodes an executable file/program into main memory.

- This is the part of the Operating System that loads the executable file on the Floppy disk/Hard disk (Secondary memory) into the Main memory and starts its execution.

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